

Question 5

Q: A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.

Theoretical Concept: Levels of Computer Vision

A surveillance system involves multiple levels:

- 1. Low-Level CV:** Image acquisition, noise reduction, and contrast enhancement.
- 2. Mid-Level CV:** Motion detection (background subtraction) and edge detection.
- 3. High-Level CV:** Object recognition (identifying a person or vehicle) and semantic understanding (detecting suspicious behavior).

OpenCV Python Solution

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```
import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for matrix operations
import matplotlib.pyplot as plt # Import Matplotlib to display images

# Step 1: Create a synthetic static background
bg = np.zeros((100, 100), dtype=np.uint8) # Create a blank black image array filled with zeros

# Step 2: Create a synthetic surveillance frame with a moving object (a square)
frame = bg.copy()
cv2.rectangle(frame, (40, 40), (60, 60), 255, -1) # Draw a filled or outlined rectangle

# Step 3: Initialize the Background Subtractor algorithm (Mid-Level CV feature)
backSub = cv2.createBackgroundSubtractorMOG2() # Create MOG2 background subtractor

# Step 4: Apply the algorithm to learn the background, then detect motion in the frame
backSub.apply(bg) # Apply background image to background subtractor to get foreground mask
motion_mask = backSub.apply(frame) # Apply frame to background subtractor to get foreground mask

# Step 5: Visualize the extracted motion mask
plt.imshow(motion_mask, cmap='gray') # Render the image array to the display
```

```
plt.title('Mid-Level CV: Extracted Motion Mask') # Add a descriptive title to the
plt.show() # Show the final plot window to the user
```

Expected Output

The output will display a completely black image with a stark white square in the center. The algorithm successfully ignored the static background and mathematically isolated the 'moving' object.

How to Execute & Submit

1. Google Colab Execution:

- Open [Google Colab](#) and create a New Notebook.
- Click the  **Copy** button on the code block above.
- Paste it into a Colab cell and press **Shift + Enter** to run.

2. Uploading to GitHub:

- In Colab, go to **File > Save a copy in GitHub**.
- Authorize your GitHub account.
- Select your Computer Vision Lab repository and click **OK** to commit the code.